



亞洲大學
ASIA UNIVERSITY

Midterm Project Report

Advanced Computer Programming

Student Name : Owenkosi Sibahle Kunene

Student ID : 114021107

Teacher : DINH-TRUNG VU

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Chapter 1 Introduction

1.1 Github

- 1) **Personal Github Account:** <https://github.com/owenkosisk06-ui>
- 2) **Group Project Repository:** <https://github.com/wangyalal/Group1>

1.2 Overview

In this assignment, I used Python and the Scrapy framework to develop a web scraping application. The program extracts repository information from my GitHub account, including URL, About, Last Updated, Languages, and Number of Commits.

I used object-oriented programming by defining an item class to structure the data. I applied web scraping techniques such as sending HTTP requests, parsing HTML using CSS selectors and XPath, and handling conditional logic for empty repositories. The program also processes and cleans the extracted data before storing it.

The final output is saved in an XML file, which organizes the scraped data in a structured format. This assignment helped me understand how to use Scrapy for real-world data extraction and how to manage and store data efficiently.

Implementation

1.1 Class 1

The `RepoItem` class is used to define the structure of the data extracted from GitHub. It acts as a container to store repository information collected by the spider.

1.1.1 Fields

- `repo_name`: Stores the name of the repository
- `url`: Stores the repository link
- `about`: Stores the repository description
- `last_updated`: Stores the last updated date
- `languages`: Stores the programming languages used
- `commits`: Stores the number of commits

1.1.2 Methods

This class does not contain custom methods because it is a data container provided by Scrapy.

1.1.3 Functions

No additional functions are defined in this class.

1.2 Class 2

GithubSpider: The `GithubSpider` class is the main component of the program. It is responsible for sending requests to GitHub, navigating repository pages, and extracting the required information.

1.3 Method/Function 1

The `parse()` method is used to process the GitHub repositories page. It extracts all repository links that belong to the user. It filters valid repository links and removes duplicates. Then, it sends requests to each repository page for further data extraction.

1.4 Method/Function 2

The `parse_repo()` method is used to extract detailed information from each repository page. It collects the repository name, URL, description, last updated time, programming languages, and number of commits.

It also includes conditional logic:

- If the repository is empty, some fields are set to `None`
- If the description is missing, the repository name is used instead

Finally, the extracted data is stored in the `RepoItem` object and exported to an XML file.

Chapter 2 Results

1.1 Result 1

The program successfully extracted repository information from my GitHub account. The extracted data includes repository name, URL, description (About), last updated date, programming languages, and number of commits.

The results were stored in an XML file (repos.xml). Each repository is represented as an item in the XML file, showing structured and organized data.

1.2 Result 2

The output of the program is saved in an XML file. The XML file contains multiple `<item>` elements, each representing a repository. The data is structured with fields such as `<repo_name>`, `<url>`, `<about>`, `<last_updated>`, `<languages>`, and `<commits>`.

Some repositories have missing descriptions or are empty, and in those cases, the program correctly handled them by assigning appropriate values such as `None` or using the repository name.

Conclusion

In this assignment, I successfully developed a web scraping program using Python and Scrapy to extract repository information from my GitHub account. The program was able to navigate web pages, extract relevant data, and store it in a structured XML format.

Through this project, I learned how to use Scrapy for web crawling, how to apply CSS selectors and XPath for data extraction, and how to handle different cases such as empty repositories and missing data. I also improved my understanding of object-oriented programming and data organization.

Overall, this assignment enhanced my practical skills in web scraping and data processing, and it provided valuable experience in building real-world applications using Python.